

MK-UNet on a 6GB GPU

Prepared for Prof. Md Mostafijur Rahman | Texas Tech University | 30 September 2026

Two independently trained models were evaluated: the official MK-UNet architecture on the authors' ClinicDB split, and the same architecture trained from scratch on Massachusetts Buildings. All values below come from saved checkpoints and actual local runs. This is a single-seed reproduction attempt, not an exact reproduction of the paper's five-run average.

Measure	ClinicDB reference experiment	Massachusetts Buildings
Training / validation / test	489 / 61 / 62 images	137 / 4 / 10 complete tiles
Input	352 x 352; scales 256, 352, 448	1500 x 1500 tiles resized to 512 x 512; four 256 x 256 patches
Optimizer / epochs	AdamW; lr = 1e-4; wd = 1e-4; 200 epochs	Same optimizer; 200 epochs; single-scale
Precision / augmentation	Mixed precision; no augmentation	FP32; no augmentation
Selected epoch	58	136
Native mean Dice, raw sigmoid	89.59%	71.95%
Native mean IoU, raw sigmoid	83.62%	56.25%

Environment and evidence

NVIDIA GeForce RTX 4050 Laptop GPU (6GB); Windows; PyTorch 2.6.0+cu124; CUDA 12.4; timm 0.9.16. The official architecture is unmodified and contains 315,566 parameters. Source revision and file SHA-256 hashes are recorded in each run's configuration and data manifest.

ClinicDB elapsed run time: 67.4 minutes; reported peak tensor allocation: 5095 MiB. Massachusetts: 15.8 minutes; 3282 MiB. Timing includes validation and test-data loading, but excludes the final test-scoring passes. Allocation includes cached dataset tensors and is not an architecture-only memory estimate.

Working code and recorded runs

How to reproduce and audit

Commands and preserved artifacts

Run `python download_data.py clinic` and `python download_data.py mass` from the archive's project root. Then run `python train_screening.py --dataset clinic --epochs 200 --batch 16 --amp` and `python train_screening.py --dataset mass --epochs 200 --batch 16`. See README for environment setup and directory conventions. Dataset downloads are excluded from the archive.

For each experiment: `config.json` records the environment and decisions; `history.jsonl` records every epoch; `best.pt` and `last.pt` preserve checkpoints; `data_manifest.json` records sample identities and hashes; `results.json` contains native-resolution aggregate and per-image results. Raw console logs accompany the epoch logs. The pilot directory records stress conditions, per-case adaptation decisions and metrics.

Protocol choices and codebase observations

The paper uses AdamW at $1e-4$, batch size 16, 200 epochs and no augmentation. The current repository script instead defaults to lr $5e-4$ with a cosine scheduler, batch size 8 and augmentation enabled. The wrapper follows the stated paper settings and omits the scheduler. Mixed precision is a declared hardware-driven change. The model is seeded, but cuDNN benchmarking is enabled; bitwise repeatability is not claimed.

Input resizing uses PIL rather than upstream OpenCV/Albumentations, and masks use the fixed >127 rule rather than upstream >20 . The official loader returns PIL image dimensions as (width, height), while the evaluator names them (height, width). The wrapper obtains the native mask shape directly, preserves its labels, resizes logits to that shape, and thresholds the original mask at 127. It also resets each multiscale input from the original batch; the current script reassigns images within the scale loop. Validation alone selects the checkpoint, and the held-out test metrics are computed after training.

Metric definitions and comparison limits

$\text{Dice} = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$; $\text{IoU} = \text{TP} / (\text{TP} + \text{FP} + \text{FN})$, with a $1e-6$ stabilizer. Primary metrics average per original image or tile, after bilinear logit resizing and sigmoid ≥ 0.5 . An all-empty prediction and all-empty target score 1. No test labels are used for threshold tuning. Massachusetts logits are stitched into a 512×512 tile before resizing to its original 1500×1500 mask.

For comparison to the code's normalization convention, native ClinicDB Dice after per-image min-max normalization is 89.59%. The paper reports 93.48% averaged over five runs [1]; our signed difference is -3.89 percentage points. Differences in precision, software, evaluator corrections and run count prevent an exact-equivalence claim. The authors' frame split may share video sequences; exact-image hash separation does not establish patient or sequence separation.

Massachusetts is a supervised transfer of the architecture to a different task. Training a new model on its labels is not evidence that a ClinicDB model generalizes zero-shot to buildings. The official tile split prevents patches from the same tile crossing partitions; neighboring tiles may still be spatially correlated.

Sources: [1], [2], [11], [12] in the proposal bibliography. Working code and recorded runs

Replication, evaluation and domain transfer

Yuvraj Verma | Screening report for Prof. Rahman | 30 September 2026

1. Why this architecture

MK-UNet is a useful test of whether a compact model can remain experimentally useful when both data and compute are limited. Its encoder combines multiple depthwise kernels with pointwise channel mixing. Channel and spatial attention refine decoder features, while grouped gates modulate skip connections. The final full-resolution output alone is optimized for binary segmentation [1,2].

The screening experiments keep those learned modules unchanged. ClinicDB tests whether the architecture can be trained under the paper's intended optimization recipe on consumer hardware. Massachusetts tests whether it remains useful when the foreground changes from a polyp to many small building footprints. Each domain has its own learned weights; the experiment transfers an architecture, not a semantic label mapping.

2. Reference experiment and replication finding

ClinicDB test measure	Observed
Paper five-run mean Dice [1]	93.48%
Single-run native Dice, min-max normalized	89.59%
Single-run native Dice, fixed sigmoid threshold	89.59%
Native mean IoU, fixed sigmoid threshold	83.62%
Checkpoint selected on validation	Epoch 58

The signed gap to the reference is -3.89 percentage points. Min-max normalization changes the current mean by less than 0.01 point, so it does not explain that gap. This attempt preserves the published optimizer, loss, channels, multiscale inputs and training duration, while changing precision and correcting evaluation details. A mismatch could reflect optimization variance, protocol drift, or evaluation differences. Without repeated runs and a harmonized evaluator, it cannot establish that the published result is wrong.

3. What changed outside medicine

Massachusetts uses all 137 training tiles, all four validation tiles and all ten held-out test tiles. Tiles are resized to 512 x 512 and divided into four non-overlapping 256 x 256 patches. The selected checkpoint achieves native mean Dice 71.95% and mean IoU 56.25%. The adaptation is supervised training from scratch, using the same network, binary structure loss, and optimizer. It uses one scale to make its compute budget explicit.

Building footprints distribute foreground across many small components. Here, held-out building masks average 18.61% foreground pixels, versus 6.85% for the polyp masks; sparsity alone cannot explain domain difficulty. Downsampling can erase narrow roof details before the model sees them; this is an input limitation, not merely a decoder limitation. Stitching four patches also exposes seam artifacts. A future resolution experiment should compare this inexpensive preprocessing with native-resolution crops on the same tile split.

Working code and recorded runs

Bottlenecks and reliability evidence

4. Execution bottlenecks and controls

Activation memory remained a constraint despite only 315,566 parameters. ClinicDB uses mixed precision and three scale passes; buildings use FP32 and one scale. Their elapsed times therefore do not measure comparable architectural efficiency.

An early diagnostic building run erased red labels through grayscale conversion, producing invalid empty-mask scores. It was stopped and restarted with red-channel decoding after every mask was checked for black/red encoding. Invalid logs are preserved but excluded from reported metrics. Filename pairing, shape checks, image hashes and native masks provide the remaining data controls.

5. Predeclared stress tests

After supervised checkpoint selection, both models were tested under fixed brightness reduction (0.65), gamma (1.8), Gaussian noise (standard deviation 0.08 on [0,1]) and Gaussian blur (sigma 1.5). These are synthetic appearance shifts, not a claim of inter-hospital or inter-sensor validation. An exploratory one-step update changes only BatchNorm affine parameters, keeps source running statistics frozen, and resets before each case. A gate accepts a candidate only if horizontal-flip Jensen-Shannon disagreement decreases by at least 1e-6 and its predicted foreground area changes by no more than five percentage points.

Domain / condition	Frozen Dice	Gated Dice	Accepted
ClinicDB / clean	89.59%	89.57%	19/62
ClinicDB / brightness_0.65	80.30%	80.32%	27/62
ClinicDB / gamma_1.8	63.78%	63.81%	17/62
ClinicDB / noise_0.08	24.11%	24.10%	26/62
ClinicDB / blur_1.5	88.89%	88.89%	21/62
Buildings / clean	71.95%	71.95%	0/10
Buildings / brightness_0.65	59.13%	59.13%	0/10
Buildings / gamma_1.8	52.50%	52.50%	0/10
Buildings / noise_0.08	5.34%	5.34%	0/10
Buildings / blur_1.5	9.64%	9.65%	1/10

Across 310 polyp decisions, ungated candidates caused eight losses >1 Dice point; the gate retained one and accepted 110 candidates. For buildings, only 1/50 was accepted: blur benefit fell from +0.526 to +0.003 points. The gate sacrificed benefit while permitting harm. This is a specified BN-affine pilot, not full Tent/EATA/SAR. Per-case records include bootstrap intervals and measured overhead.

6. Interpretation and next experiment

Next, compare native-resolution crops with this low-resolution baseline, repeat seeds, and measure harmful acceptance and useful rejection at matched computation. These runs establish execution feasibility and expose reliability failures; they do not establish clinical usefulness or superiority over current adaptation methods.

References: [1-12] are listed in the accompanying proposal. Working code and recorded runs

Learning behavior and failure cases

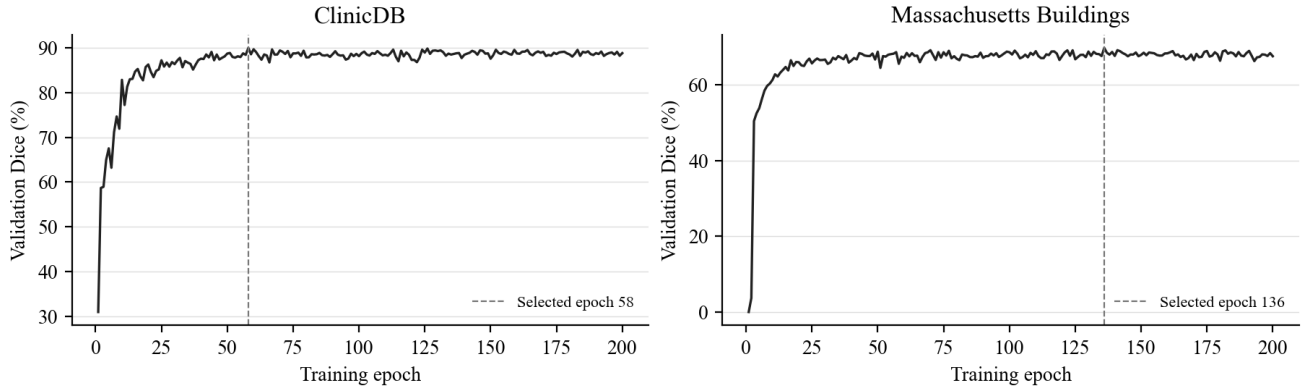


Figure 1. Validation trajectories over the complete 200-epoch runs. Dashed lines mark the selected checkpoints. The source and target selection metrics have different normalization conventions and should not be compared as equivalent scores.

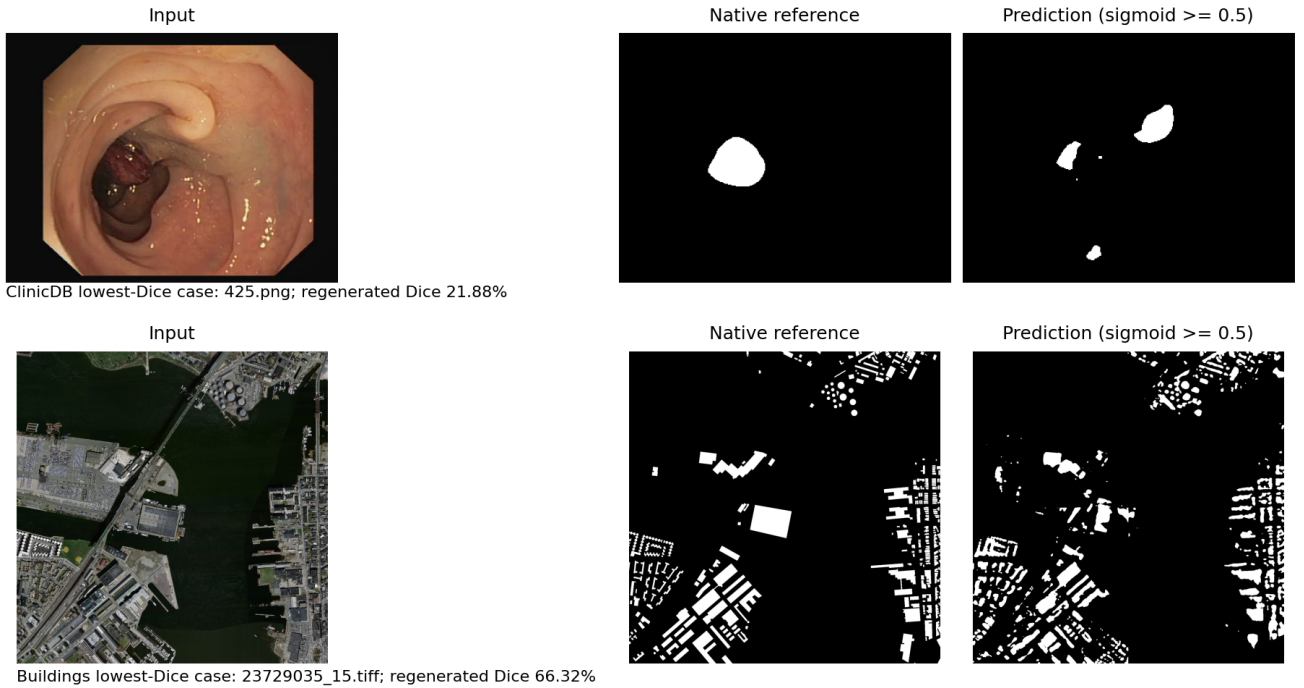


Figure 2. Lowest native fixed-threshold Dice test case in each domain, selected explicitly from the saved per-case results. Predictions were regenerated from best.pt and their Dice recomputed; small inference-rounding differences are recorded in the code archive. Display images are scaled to fit the page; the metric uses the original mask resolution.

These examples are evidence of failure modes in the tested pipeline, not a diagnosis of their cause. The proposed resolution and boundary ablations would test whether lost input detail, patch seams or adaptation decisions explain them.

Budgeted test-time adaptation with rollback for lightweight segmentation

Yuvraj Verma | Proposed first project with Prof. Md Mostafijur Rahman | Texas Tech University

Research pillars: continuous adaptation and reliability; scalable AI for constrained hardware. **Initial settings:** polyp segmentation and aerial building segmentation, trained as separate tasks. **Available platform:** one 6GB RTX 4050 laptop GPU.

1. Motivation and significance

An efficient segmentation model can still fail when illumination, blur, noise or acquisition conditions change. A deployment that repeatedly updates from unlabeled inputs may reduce some of those errors, but it can also become confidently wrong. The first project would ask whether update decisions can be constrained and audited as carefully as the model itself, without introducing a large auxiliary network.

For medical images, a missed or distorted lesion boundary can make a high aggregate overlap misleading. For remote sensing, missing many small roofs can bias an inventory even when large buildings dominate the pixel score. These are different semantic tasks, but both expose a practical engineering question: when should a compact model spend additional computation adapting, and when should it retain its original prediction? This proposal addresses that shared decision, rather than assuming a model trained on one task can recognize the other.

2. Research question and falsifiable hypotheses

Question: under a fixed, measured adaptation budget, can a lightweight accept-or-rollback rule reduce harmful updates while preserving useful gains on shifted inputs? The rule will be evaluated against both frozen inference and the same candidate update without gating.

H1: gating reduces the fraction of cases with a Dice loss exceeding one percentage point relative to the frozen model, compared with an ungated update, at matched candidate-computation budgets. **H2:** an input-level trigger can avoid unnecessary updates on clean cases without losing most of the ungated shifted-case benefit. **H3:** flip consistency alone is an insufficient acceptance test; adding a foreground-change or boundary-localized condition improves its ability to reject damaging updates.

The one-point threshold is a predeclared experimental definition, not a clinical tolerance. Each hypothesis can fail: a gate may reject beneficial candidates, accept consistently wrong masks, or consume more time than the update saves. The project will report those outcomes rather than define success solely through average Dice.

3. Proposed scope

The initial study uses binary segmentation and a small architecture with source-trained BatchNorm. It does not attempt full-model foundation-model tuning, clinical reasoning, multi-organ 3D analysis or cross-task zero-shot transfer. A later brain-MRI study would require an appropriately labeled dataset, subject-level partitions and 3D-aware evaluation; it will not be inferred from these 2D screening results.

Preliminary source and target training are complete; measured results and the small adaptation pilot appear on page 5. The full trigger, boundary localization and repeated-seed study remain proposed work.

Positioning the project and defining the gap

4. What existing methods already address

Tent [3] adapts normalization-related parameters through predictive entropy. EATA [4] selects useful samples and constrains forgetting; CoTTA [5] combines teacher-based predictions with restoration; SAR [6] examines stability under changing test conditions. These works already establish that selective updates, restoration and reliability matter. A rollback mechanism alone is therefore not a novelty claim.

Medical-segmentation-specific work strengthens that caution. On-the-Fly adaptation [7] addresses test distribution differences in medical images. Chen et al. [8] refine foundation-model representations without model parameter updates. MedSeg-TTA [9] compares adaptation paradigms across multiple modalities and shows that their behavior depends on the shift. Recent memory-supported training-free adaptation [10] is another relevant comparison. Methods [8,10] rely on foundation-model components, so they are contextual comparators rather than drop-in MK-UNet algorithms. The primary causal comparison will use the same MK-UNet checkpoint and compute accounting; any foundation-model experiment will report its distinct backbone, prompts and resource budget.

Prior direction	What this study would test
Normalization / entropy adaptation [3]	Does a very small model still make useful candidate updates when its running statistics stay frozen?
Selective / stable adaptation [4-6]	Does a decision rule reduce harmful segmentation updates at a measured compute budget?
Medical / foundation-model approaches [7,8,10]	When are parameter-free or input-level corrections a better use of the available budget?
Cross-modality benchmarking [9]	Can failures be explained by object size, boundaries and shift type instead of a pooled mean?

5. Specific gap and contribution target

The proposed contribution is a controlled segmentation study of the tradeoff between update harm, useful adaptation and total inference cost on a sub-million-parameter backbone. The experiment should make the gate itself a measurable object: false acceptance of harmful updates, false rejection of beneficial updates, spatial localization of unstable predictions, and added latency. Whether this remains a publishable gap will be reassessed against the detailed protocols of [7-10] before claiming novelty.

The project is deliberately narrower than proposing another large adaptable backbone. It should produce an auditable decision policy, an ablation suite, and evidence about when that policy cannot infer correctness from unlabeled observations. If the simple gate fails, identifying the failure boundary remains a useful result and will guide a smaller follow-up.

6. Mathematical problem

Let $f(\theta, x)$ return binary foreground probabilities; θ_0 is a source-trained snapshot. A test case x arrives without its mask y . A candidate θ_1 is formed from x using a restricted parameter subset. A policy g observes unlabeled predictions, input statistics and computational budget, then chooses θ_1 or θ_0 . Ground-truth y is available only to the offline evaluator. The objective is to improve held-out overlap while limiting update harm and cost; label-free proxy improvement is not equivalent to segmentation improvement.

A small candidate update with an explicit decision rule

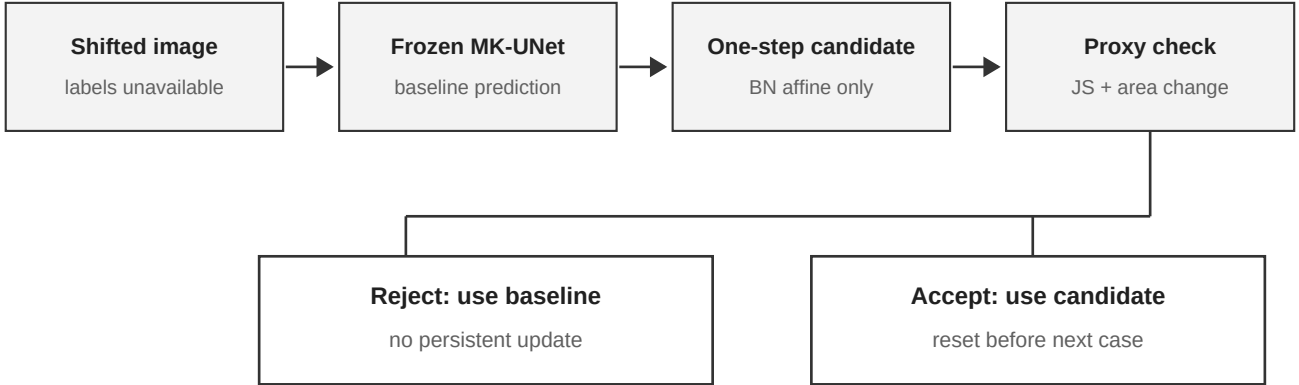


Figure 1. Initial episodic framework. The candidate and acceptance test use no target labels. Accepted weights apply only to the current case; the next case starts from the source snapshot. The proposed input trigger and boundary map are additions to this pilot.

7. Candidate update

Freeze convolutional weights and source BatchNorm running statistics; adapt only BN scale and offset. For binary probabilities p , entropy is $h(p) = -p \log p - (1-p) \log(1-p)$. Compute mean entropy separately within the baseline-predicted foreground and background, then average non-empty strata. This prevents the large background region from dominating the loss, but does not guarantee that either predicted stratum is correct.

The screening pilot uses one SGD step at learning rate $1e-3$. The full study will compare one-step and three-step candidates, a source-snapshot parameter penalty, and a consistency term. For multiple steps, the proposed objective is $L = H_{\text{balanced}} + \lambda_{\text{consistency}} JS(p(x), T^{-1} p(Tx)) + \lambda_{\text{anchor}} \|\phi - \phi_0\|^2$, where ϕ contains adaptable BN affine parameters. Hyperparameters are selected using source validation and prespecified validation corruptions, then locked before held-out testing.

8. Detection, localization and acceptance

First compute a frozen prediction and a transformed prediction. Pixelwise Bernoulli Jensen-Shannon disagreement gives an instability map rather than an explanation of model reasoning. The proposed trigger tests whether its upper quantile exceeds a validation-calibrated limit. A boundary band, derived from the baseline mask, separates edge instability from changes in interiors. Regions can be shown to a reviewer, but cannot be labeled as actual errors without evidence.

After the candidate update, recompute the proxy and predicted foreground area. The pilot accepts only if mean JS decreases by at least $1e-6$ and foreground area changes by at most 0.05 of the input area. The full rule will compare this with boundary-weighted disagreement, relative foreground-area bounds, bounded parameter change and an empty-mask check. An absolute five-point area bound can permit a large relative loss when foreground is sparse. Legitimate appearance or object-size changes can violate these bounds; rejection is conservative behavior, not proof of a bad candidate.

9. Budget and rollback

A hard cap on forward and backward passes sets the budget. Report all candidate and gate costs, including rejected updates. Episodic reset prevents persistent drift between cases; it cannot prevent harm in the current mask. Continual adaptation would require a separate recovery study.

Protocols designed to expose failure

10. Data and partitioning

Stage A starts from the ClinicDB replication checkpoint and the separately trained Massachusetts checkpoint. Use the recorded source frame split for the replication comparison, but use sequence- or patient-disjoint data for later generalization conclusions wherever identifiers allow it. For Massachusetts, retain the original train/validation/test tile split, and assess spatial dependence before interpreting intervals. All crops from a parent tile stay in one partition.

Stage B adds independent polyp acquisition data such as Kvasir-SEG after verifying access and image overlap, and a second geographical or sensor domain for building segmentation. Target labels are reserved for offline scoring. The source-only and supervised target-trained baselines answer different questions and will be reported separately. A new benchmark chosen after inspecting failures requires a new, explicitly labeled evaluation phase.

11. Comparator suite and ablations

Core controls are frozen inference, an identity-update numerical control, flip-based test-time augmentation, inexpensive input normalization, ungated BN-affine adaptation, and the gated version of the same candidate. These establish whether the gate offers value beyond averaging predictions or correcting inputs. Add compatible implementations of Tent, EATA and SAR with clearly documented segmentation adaptations. Where feasible, use the standardized medical protocols of [9] and a parameter-free alternative [8,10]; exclude methods only with a stated resource or architecture reason.

Ablation	Question
No trigger; always form a candidate	Does conditional computation actually save time?
No gate; accept every candidate	How much harm is caused by adaptation itself?
JS only / area only / both	Which signal rejects harmful cases, and which rejects useful ones?
Full-image / boundary-localized proxy	Does localization add value beyond mean disagreement?
1 vs 3 steps; BN affine vs head	Does extra update capacity justify cost and risk?
256/352 inputs vs native crops	Is loss of small structures caused mainly by downsampling?

12. Outcomes and analysis

Report native per-image Dice and IoU, boundary F1 with a documented pixel or physical tolerance, and results stratified by foreground area. For each method, report the rate of Dice losses greater than one point, the mean gain among accepted candidates, harmful-acceptance rate, useful-rejection rate, and accept frequency. Use paired differences against the frozen baseline. Report exact counts, per-case plots and worst failures.

Repeat training with at least three fixed seeds in the full project and use paired bootstrap intervals at the patient/sequence or parent-tile level. The screening pilot uses a single seed and simpler image/tile bootstrap intervals. Measure latency with GPU synchronization and warm-up; report p50/p95 total latency, forward/backward counts, peak allocated and reserved VRAM, and rejection overhead. Synthetic corruptions support a stress-test conclusion, not a clinical or sensor-transfer claim.

Completed runs and a deliberately small pilot

Completed experiment	Native raw Dice	Native raw IoU
ClinicDB; 62 test images	89.59%	83.62%
Buildings; 10 test tiles	71.95%	56.25%

The official architecture has 315,566 parameters. ClinicDB used 200 epochs and mixed precision; Massachusetts used 200 epochs in FP32 at a lower input resolution. These experiments demonstrate execution feasibility, not an adaptation advantage. The source min-max-normalized Dice is 89.59%, compared with the paper's five-run mean of 93.48%. The execution summary documents all material differences.

13. What the adaptation pilot actually tested

The pilot updates 10,872 BN affine parameters for one step, with source statistics frozen, then performs the JS-and-area acceptance test. It resets weights for every image or complete four-patch tile. No additional candidate-learning rate or gate threshold was tuned on the test outcomes. All stress transformations were fixed before the pilot ran.

Condition	Clinic frozen / gated	Buildings frozen / gated
clean	89.59 / 89.57	71.95 / 71.95
brightness_0.65	80.30 / 80.32	59.13 / 59.13
gamma_1.8	63.78 / 63.81	52.50 / 52.50
noise_0.08	24.11 / 24.10	5.34 / 5.34
blur_1.5	88.89 / 88.89	9.64 / 9.65

Values are mean native Dice percentages. Mean candidate overhead was 46.5 ms per polyp image and 56.2 ms per building tile, above the two-forward frozen baseline. Only forward passes were warmed up, so these means include the first adaptation-call cold start. They exclude reset, input corruption and offline native scoring. Per-case records include ungated results, gate decisions and paired bootstrap intervals. The pilot has no input trigger, no learned risk predictor, no boundary-localized gate, and no multi-seed comparison. Those components are proposed; their benefits are not represented as measured.

14. Limits that affect interpretation

Only 62 source images and ten target tiles contribute to these held-out summaries. Nearby source video frames and neighboring aerial tiles can be correlated, so naive image/tile confidence intervals may be too narrow. Downsampled building inputs can remove small objects. Neither dataset establishes clinical usefulness, real edge-device throughput, neuroimaging performance or robustness across institutions. A decrease in flip disagreement cannot certify that a prediction is correct.

The noise condition reduced frozen Dice to 24.11% for polyps and 5.34% for buildings. Blur reduced building Dice to 9.64%, while polyp Dice stayed at 88.89%. The gate accepted 110/310 polyp candidates, including one loss >1 point, but only 1/50 building candidates. This demonstrates a harm-versus-benefit tradeoff, not a successful reliability solution. If harmful candidates remain accepted, the next step will test boundary localization or avoid adaptation on those cases. If the candidate rarely changes predictions, the result motivates a capacity-and-step ablation, not a claim that rollback solved reliability.

Working code and recorded runs

A feasible first research project

15. Milestones and expected outcomes

Period	Concrete milestone and decision
Weeks 1-2	Harmonize reference evaluation; repeat seeds; confirm source/target access and group identities. Release reproducible manifests and leakage checks.
Weeks 3-4	Establish frozen, input-correction, TTA-averaging and ungated-update baselines. Lock validation-derived trigger and gate settings.
Weeks 5-6	Run gate ablations and object-size/boundary analyses. Measure harmful acceptance, useful rejection and synchronized latency.
Weeks 7-8	Test an independent acquisition/geographical domain; stress clean-to-shift-to-clean streams separately from episodic tests.
Weeks 9-10	Repeat key findings; inspect failure cases; package code and write the study. Continue only where gains exceed uncertainty and measured overhead is justified.

Expected artifacts are a compact decision policy, a reproducible evaluation harness, an interpretable map of prediction instability, and evidence about the circumstances in which adaptation is useful or harmful. Publication venue would follow the strength and scope of the results. A negative result would retain value if it identifies a reproducible failure mode and establishes a reliable fallback; a clinical deployment claim would require additional evidence beyond this project.

The work fits experience with constrained-GPU adaptation and error analysis, while adding segmentation, partition-aware evaluation and adaptation decision-making. It aligns with Prof. Rahman's efficiency and reliability pillars without requiring a large new backbone or assuming access to a hospital dataset.

References and working code

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Working code and recorded runs

Assistance note. AI tools supported code development, literature organization and initial drafting. All reported measurements come from executed experiments and saved evidence.